Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and use their own software and AI to make them faster and with fewer skilled people than older competitors. They earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below cover why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production are lasting and others can reverse. The lasting ones are what we should build a thesis on.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a big buyer to approve a new supplier. None of these goes away with an election or a budget cycle.
   - **Temporary:** tariffs, the current surge in munitions spending and interest rates. Tariffs are set by proclamation and can change overnight. Munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** we should favor companies that still work if tariffs fall and the munitions surge slows. That means owning approved capacity that buyers need for a decade, and avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets (Mining.com). These magnets go into drones, missiles, motors and robots.
   - The US is fully import-reliant for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - This is why buyers and the government will pay for a non-Chinese supply even when it costs more. It also means the plants that fix this problem are expensive and slow to build.
3. **Defense demand and government money are real, but much of the money is conditional and slow.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the past decade's average. About 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years, with advance payments. That is new and helpful, though each year still needs Congress to approve the money (FY26 NDAA).
   - Government now acts as lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). Governments worldwide put about $65B into minerals in 2025 (IEA). These loans take a long time to close and come with conditions.
4. **The labor gap is lasting, and it is driving owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of them may go unfilled (Deloitte).
   - Of 17,156 US machine shops, 83% have fewer than 20 employees (Census). Many owners are near retirement with no one to take over. That creates a supply of certified shops for sale at reasonable prices.

- Fewer skilled people means a plant that needs fewer of them can grow where others cannot. That is the opening for software and automation.

5. **Capex before contracts is the central risk. A factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A big defense contractor, often called a prime, won't approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes about nine months to two years: one prime told us 9-12 months in an expert call, and another said one to two years.
   - One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says capex runs about two years ahead of revenue (claim). Process plants need about 80-85% use of capacity to make money (FTI Consulting). A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
   - This hurts venture returns. Equity spent on a building earns building-like returns and dilutes founders before the first volume order arrives.
   - The winners shift who pays for the build. Options include government loans and grants, a buyer that agrees to purchase output before the plant is built (the Pentagon agreed to buy all of one magnet plant's output for 10 years, with a price floor), a customer that hosts the factory inside its own plant (one 7-year deal), bank debt once revenue exists, leased buildings, franchising, and buying plants that are already certified. One roll-up founder said they buy shops at about seven times annual earnings (claim).
6. **Buyers pay for speed and spare capacity. Few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a defense prime said in an expert call. An engineer there would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it farms out to partner shops (company filings). Approved incumbents earn well too: one engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Aerospace programs can run about 50 years, so an approval keeps paying.
   - Selling software or machines to factories and mines has been hard. One 3D-printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). An AI-run machining startup in Germany went insolvent in August 2026. Buyers cap each supplier at about a quarter to a third of their work, so even a winner grows one share at a time.
7. **AI's edge today is in the work around the machine and in processes that run nonstop. Nobody has shown yet that it makes each unit cheaper as volume grows.**
   - Around the machine means quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut assembly-instruction writing from about 10 hours to 15 minutes (claim). A former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves are underused. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target. European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Better planning and control software closes that gap.
   - In a plant that runs nonstop, like a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. A head of strategy at an aircraft maker put net savings from automation, after paying for the machines, at 20-30% at best. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case rests on speed and yield more than headcount.
8. **The key risk is that incumbents automate and buy faster than new companies can grow.**

- The primes decide who is approved and how much work each supplier gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
- Incumbents buy instead of build. Miners back startups through their venture arms, one paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, one with plans for up to $1B.
- There is still room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that base is small, old and short of people, which is where new companies can win.

2. Interesting categories

This section follows the categories in our primer. Each starts with our call on whether we can invest today, then why, how the business works, and the companies the primer names there.

A. Raw Materials

Stage A: Locating & finding minerals and mines

Investability: Economics don't work for us.

- **Why:** AI can help explorers choose where to drill, but the payoff comes too late and too rarely for venture money. Mines average 15.7 years from discovery to production, and 19.1 years for US mines opened in 2020-24. One explorer puts the industry's success rate at about 0.5% per project (claim). Its CEO says about 40 deposits are found a year and about a fifth become mines (claim). Most finds never earn anything.
- **How the business works:** The explorer owns mining claims and drill rigs and is paid only when it proves a deposit and sells it to a miner, for example for about 5% of the metal in the ground plus a royalty (claim). Until then it spends on drilling and geologists with no revenue. The miners who buy deposits also run their own exploration and fund many startups through venture arms, so the field is crowded. Software sold to miners on its own rarely sticks; one explorer ran about 80 pilots that did not turn into repeat business.

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

Investability: Selective. Only where government debt pays for most of the plant and a buyer has signed for the output.

- **Why:** Chinese refineries take about 6 months to commission against 2 to 4 years in the West. Defense and auto buyers need non-Chinese metal and magnets, and the government is lending and signing long purchase deals to get plants built.
- **How the business works:** Miners run the mine and sell metal such as copper at market prices; one startup says its restarted Utah copper mine runs about half autonomously (claim). Refiners buy ore concentrate or scrap and sell oxides, metals or magnets under long contracts. Both own their plants and spend hundreds of millions of dollars before revenue: about 5 years to build and 3 to 5 to ramp after permits (claim). The deals that work push that cost onto others, such as a $620M government loan to one magnet maker, or a 10-year price floor and a promise that all output of an incumbent's next magnet plant will be bought. Without a floor, a US producer takes prices set in China. The largest US rare-earth producer lost $85.9M on $224.4M of revenue in 2025.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

Investability: Economics don't work for us.

- **Why:** This is a crowded business run by established incumbents, and there is little technology to set a new company apart. The largest US distributor is gaining share without AI. Tariffs help US metal, but the margin stays with the incumbents.
- **How the business works:** The distributor buys metal from mills, holds it in stock, cuts it to size and sells it to shops and primes. It earns a spread on the metal plus cutting fees, and much of its cash sits in inventory. Online ordering with instant quotes already exists, so a new distributor has little to sell beyond price and stock.

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

Investability: Investable now.

- **Why:** Fabrication is cutting, bending and welding metal into frames and enclosures, and assembling parts like wire harnesses, the bundles of wires that carry power and signals through an aircraft or missile. Plants cost far less than in precision work, and the best companies already have revenue. One online fabricator handles about 32,000 orders a month, is growing about 100% a year and says it has been profitable for eight years (claim). One harness maker claims it cut lead time from six months to two weeks (claim).
- **How the business works:** Customers send a design and get parts or assemblies back in days. Most companies own their plants; one leased a second plant and opened it within months. Some route work to partner shops and keep less of each sale: one listed parts maker earns a 49% gross margin in its own plants and 31% on work sent to partners (company filings). Speed is the edge, since one founder says he beats China on price on only about 20-25% of his work (claim). The risk is that customers outgrow them and buy their own machines.

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision & Defense Hardware

Investability: Investable now. We favor models where someone else pays for most of the machines.

- **Why:** Precision parts are metal or composite parts made to very tight limits, by machining, 3D printing, casting or molding, for missiles, jets, satellites and submarines. The shops that make them are small and slow, and primes, which earn 7-12% on cost-plus contracts, outsource much of the work. A supplier that delivers fast earns a premium and, once approved, keeps the work for programs that can run 50 years.
- **How the business works:** Companies sell parts under purchase orders or multi-year contracts, and the factory model decides who pays for the machines. Owned factories built from scratch keep the whole margin and all the data but pay for the build before approval: one parts maker committed about $200M to a single plant, and a former employee said a six-machine first plant cost "well north of $10 million." Buying certified shops from retiring owners is cheaper, since the machines and approvals already exist; roll-ups buy at about seven times annual earnings (claim). In a franchised network, franchisees buy the machines while the company supplies software, orders and quality rules; one claims 14 factories, 4 owned and 10 franchised (claim). Cells hosted inside a buyer's plant share the cost with the buyer; one 7-year deal puts cells inside a prime's missile plant. The last two keep less of each sale but need far less capital. New plants need AS9100 certification, the aerospace quality standard, which typically takes 6-12 months, then each part's approval at the buyer.

Example companies: Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

Investability: Investable now, selectively. The older lab has drug-company clients and its own lab running; the newest has funding but no public customers yet.

- **Why:** We limit chemicals to automated labs that make new molecules for drug programs. Drug companies send most of this work offshore, largely to China. BIOSECURE, a law passed in December 2025, limits use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). Traction is real but early. One lab has run its first automated lab since June 2025, has 152 staff, and says it works with eight of the top 20 drug companies (claim, SCI); a trade publication reported 10 big pharma clients (Citeline). Its one deal with public terms pays through milestones and royalties, and it has not disclosed revenue. The newest lab launched in October 2026 and works with a small number of design partners.
- **How the business works:** The lab makes small batches of new molecules for drug teams, at milligram-to-gram scale, and does not develop its own drugs. It is paid per compound delivered, or through milestone payments and royalties if a molecule moves forward. It owns the lab, software and reaction data. Labs cost tens of millions; one is part-funding an £89.9M expansion with about £22M of UK and Scottish grants. US chemists cost several times Chinese rates, so automation has to bring unit cost down.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Producers

Investability: Selective. Invest when a buyer is already qualifying the material.

- **Why:** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type. Data centers used 415 TWh of power in 2024 and are on track for about 945 TWh by 2030, with cooling taking 7% to over 30% of it. Both create buyers for a new or non-Chinese material, and the company that designs and makes it keeps the margin.
- **How the business works:** These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals. Some own the plant; one claims its AI carries a material from 1 kg a day in the lab to 1,000 kg a day in production (claim). Others use a contract maker, which cuts capex. One says its cooling fluid took about 3 years to reach market, against roughly 10 years and $100M for a new material (claim). Customer qualification takes one to five years. Owned cathode plants cost hundreds of millions and compete on Chinese prices, and an established incumbent's battery-materials unit lost money on €436M of revenue.

Example companies: Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

Investability: Longer time horizon. More of an experiment today.

- **Why:** AI and robotic labs can test far more candidates than people can. One lab made about 1,200 alloys in five or six months (claim), and an independent university test found one of its alloys beat a standard aerospace alloy. They do not yet sell material at volume, and finding a material faster does not shorten the years customers take to qualify it.
- **How the business works:** Labs earn today from paid R&D for partners, licensing and royalties, software licenses and government research awards, which run around $1M. Their path to making material usually runs through a partner that scales it up and sells it, so the lab earns a fee or royalty unless it later builds its own production. Valuations rest on the team, from a $300M seed to one reported at about $7.5B. One investor says materials has had "no AlphaFold moment."

Example companies: Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

Enablers

Investability: Investable now for plant software paid on results. Longer time horizon for licensed factory software and robots.

- **Why:** Process plants run at about 74% of capacity in Europe against the 80-85% they need (FTI Consulting), and in a sold-out plant each extra point of yield is close to pure profit. Software that runs the plant and is paid on the output it adds fits an enterprise software fund. Most other tools here are still early, and the US installed 38,428 industrial robots in 2025 against 354,226 in China.
- **How the business works:** Plant-control software connects to existing controls, adjusts setpoints such as a temperature or flow rate itself, and one startup says it is paid on the production it adds (claim). There is no factory to build. Licensed factory software, such as tools that write assembly instructions (one claims 10 hours down to 15 minutes, claim), is sold by subscription, and incumbent automation vendors bundle software with their hardware. Robot vendors sell or rent robots, but robot-learning success is 59-66% on short tasks against the 99.9% plus a factory needs.

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson